Exercise 1: Emergency Dispatch Record Search

Execution Log (Complete, Unedited Program Output)

Seed: 3
ID Sum: 18
Name Length: 6
Artist Length: 5
Artist Sum: 61

Original Dataset: [18, 25, 32, 39, 46, 53, 60, 67, 74, 81, 88, 95, 102, 109, 116]
Rotation Point: 3
Rotated Dataset: [39, 46, 53, 60, 67, 74, 81, 88, 95, 102, 109, 116, 18, 25, 32]
Target Incident: 46

Target found at index 1
Rotated Binary Search comparisons: 3
Linear Search comparisons: 2
Time Complexity: Rotated Binary Search O(log n), Linear Search O(n)
Space Complexity: O(1)

Assessment Data

#	Field	Entry
1	Original Dataset	[18, 25, 32, 39, 46, 53, 60, 67, 74, 81, 88, 95, 102, 109, 116]
2	Rotated Dataset	Rotation point: 3 (SEED_DIGIT % 15) [39, 46, 53, 60, 67, 74, 81, 88, 95, 102, 109, 116, 18, 25, 32]
3	Target Incident	46 (which is found at index 1 of the rotated dataset)
4	Comparison Counts	Rotated Binary Search: 3 comparisons
5	Linear Search comparison count	2 comparisons
6	Time and space complexity	Rotated Binary Search: O(log n) time, O(1) space Linear Search: O(n) time, O(1) space

7	Final Output	Target found at index 1 Rotated Binary Search comparisons: 3 Linear Search comparisons: 2 Time Complexity: Rotated Binary Search O(log n), Linear Search O(n) Space Complexity: O(1)

Exercise 2: University Enrollment Records

Execution Log (Complete, Unedited Program Output)

ID Sum: 18
Name Length: 6
Artist Length: 5
Artist Sum: 61

Department A: [30, 48, 66, 84, 102]
Department B: [50, 56, 62, 68, 74]
Consolidated: [30, 48, 50, 56, 62, 66, 68, 74, 84, 102]
Records before: 5 + 5 = 10
Records after: 10
Comparisons: 8
Ordered correctly: True
Time Complexity: O(n + m)
Space Complexity: O(n + m)

Assessment Data

#	Field	Entry
1	Department A dataset	[30, 48, 66, 84, 102]
2	Department B dataset	[50, 56, 62, 68, 74]
3	Consolidated dataset	[30, 48, 50, 56, 62, 66, 68, 74, 84, 102
4	Number of records before and after consolidation	Before: 5 + 5 = 10 After: 10
5	Comparison count	8

6	Ordering verification	Verified ascending (Ordered correctly: True)
7	Time and space complexity	O(n + m) time, O(n + m) space
8	Final Output	Department A: [30, 48, 66, 84, 102] Department B: [50, 56, 62, 68, 74] Consolidated: [30, 48, 50, 56, 62, 66, 68, 74, 84, 102] Records before: 5 + 5 = 10 Records after: 10 Comparisons: 8 Ordered correctly: True Time Complexity: O(n + m) Space Complexity: O(n + m)

Exercise 3: E-Commerce Order Processing

Execution Log (Complete, Unedited Program Output)

Seed: 3
ID Sum: 18
Name Length: 6
Artist Length: 5
Artist Sum: 61

Generated Orders: [54, 33, 42, 42, 17, 89, 31, 56]

Sorted input: [17, 31, 33, 42, 42, 54, 56, 89]

Algorithm	Comparisons	Movements	Time (s)	Verified
Bubble Sort	7	0	0.00000420	True
Selection Sort	28	0	0.00000386	True
Insertion Sort	7	0	0.00000202	True

Reverse input: [89, 56, 54, 42, 42, 33, 31, 17]

Algorithm	Comparisons	Movements	Time (s)	Verified
Bubble Sort	28	27	0.00000644	True
Selection Sort	28	3	0.00000354	True
Insertion Sort	28	27	0.00000420	True

Random input: [54, 33, 42, 42, 17, 89, 31, 56]

Algorithm	Comparisons	Movements	Time (s)	Verified

Bubble Sort	27	13	0.00000436	True
Selection Sort	28	6	0.00000324	True
Insertion Sort	18	13	0.00000268	True

Assessment Data

#	Field	Entry
1	Generated order dataset	[54, 33, 42, 42, 17, 89, 31, 56]
2	Sorted, reverse, and random inputs	Sorted: [17, 31, 33, 42, 42, 54, 56, 89] Reverse: [89, 56, 54, 42, 42, 33, 31, 17] Random: [54, 33, 42, 42, 17, 89, 31, 56]
3	Comparison counts	Bubble Sorted: 7 Reverse: 28 Random: 27 Selection Sorted: 28 Reverse: 28 Random: 28 Insertion Sorted: 7 Reverse: 28 Random: 18
4	Movement counts	Bubble Sorted: 0 Reverse: 27 Random: 13 Selection Sorted: 0 Reverse: 3 Random: 6 Insertion Sorted: 0 Reverse: 27 Random: 13
5	Execution times	Bubble Sorted: 0.00000420 s Reverse: 0.00000644 s Random: 0.00000436 s Selection Sorted: 0.00000386 s

		Reverse: 0.00000354 s Random: 0.00000324 s Insertion Sorted: 0.00000202 s Reverse: 0.00000420 s Random: 0.00000268 s
6	Sorting verification	All three algorithms returned a correctly sorted list for all three input conditions (Verified: True in all 9 runs).
7	Big O analysis	All three are O(n2) in the worst case. Bubble Sort (with early exit) and Insertion Sort are O(n) in the best case (sorted input). Selection Sort is O(n2) in every case. Space complexity is O(1) auxiliary for the sorting itself.
8	Best-, average-, and worst-case observations	Best case: sorted input (Bubble and Insertion: 7 comparisons, 0 movements). Worst case: reverse input (Bubble and Insertion: 28 comparisons, 27 movements). Average case: random input falls in between (Bubble 27/13, Selection 28/6, Insertion 18/13). Selection Sort always makes 28 comparisons.
9	Final Output	Generated Orders: [54, 33, 42, 42, 17, 89, 31, 56] Sorted input: [17, 31, 33, 42, 42, 54, 56, 89] Algorithm Comparisons Movements Time (s) Verified Bubble Sort 7 0 0.00000420 True Selection Sort 28 0 0.00000386 True Insertion Sort 7 0 0.00000202 True Reverse input: [89, 56, 54, 42, 42, 33, 31, 17] Algorithm Comparisons Movements Time (s) Verified

		Bubble Sort 28 27 0.00000644 True Selection Sort 28 3 0.00000354 True Insertion Sort 28 27 0.00000420 True Random input: [54, 33, 42, 42, 17, 89, 31, 56] Algorithm Comparisons Movements Time (s) Verified Bubble Sort 27 13 0.00000436 True Selection Sort 28 6 0.00000324 True Insertion Sort 18 13 0.00000268 True